MemLabs (Trend, Reversion, Regime)

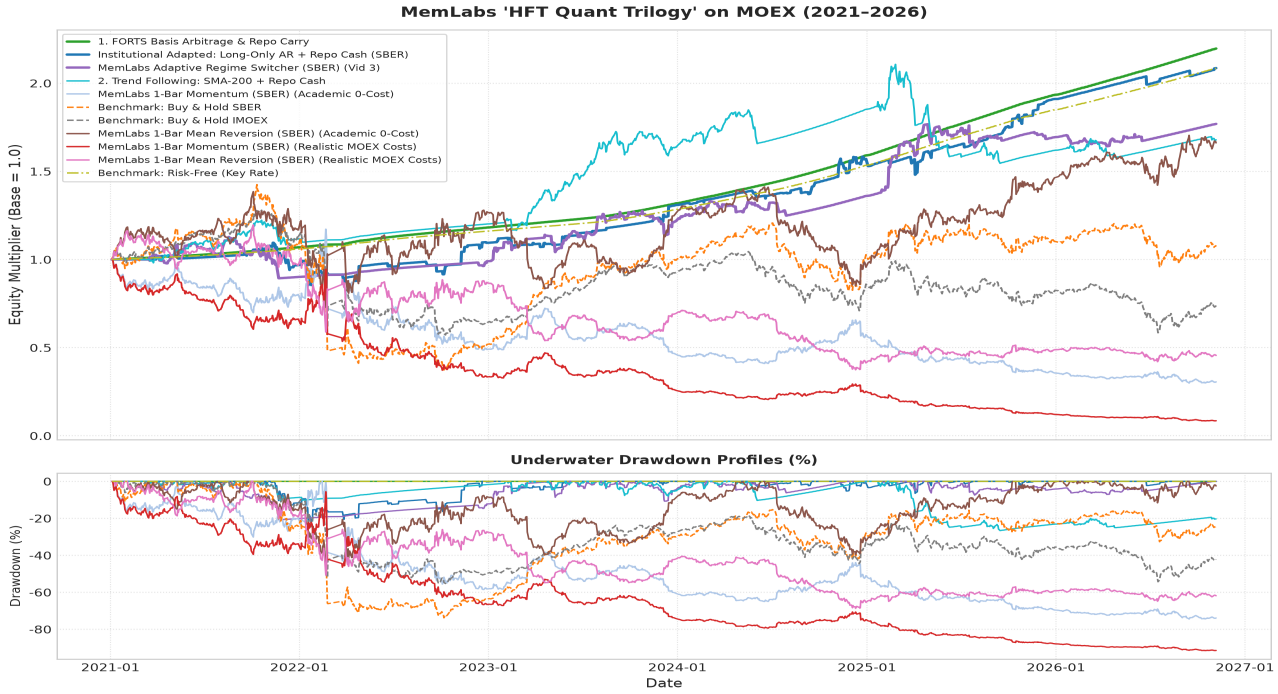
3 -HFT , 2021 2026 .
€ •

, ƒ : „ … …† -HFT
MemLabs: Trend Following , Mean Reversion How to handle Regime Changes . ‡
ˆ • ‰ (2021 2026, 1468)
, Š ‹ (, …ƒ Š + 4.0% = 25%) …†
‰ $t+1$ Š …:
1. 1-Bar Momentum (Trend Following) (-34.38% CAGR, MaxDD -91.5%) -
Œ …† …Š Œ .
2. 1-Bar Mean Reversion (+9.09%), ‰ Š
(-12.59%) - ‹ 25% .
3. • Œ Š ‰ (Regime Changes) : …ƒ …†
/ Ž„ … CAGR +10.27% -20.80%,
Œ …† Long-Only AR • CAGR +13.42%.

1. MemLabs

/ MemLabs	CAGR		Sharpe (vs)	Sortino	Max Drawdown	Calm ar
1. FORTS Basis Arbitrage & Repo Carry	14.43%	0.33%	+2.76	0.00	0.00%	>10.0
2. • Œ …† : Long-Only AR + Repo Cash	13.42%	12.82%	+0.06	0.60	-22.02%	0.61
3. Adaptive Regime Switcher (SBER) (Vid 3)	10.27%	11.60%	-0.19	0.76	-20.80%	0.49
4. Trend Following: SMA-200 + Repo Cash	9.28%	10.45%	-0.30	0.84	-26.64%	0.35
5. 1-Bar Mean Reversion (0-Cost Academic)	9.09%	33.79%	+0.05	0.55	-47.44%	0.19
Benchmark: Buy & Hold SBER	1.34%	33.82%	-0.15	0.24	-73.81%	0.02
Benchmark: Buy & Hold IMOEX	-5.05%	27.93%	-0.49	-0.04	-55.29%	-0.09
Benchmark: Risk-Free (, …ƒ Š)	13.41%	0.33%	0.00	0.00	0.00%	N/A
6. 1-Bar Mean Reversion (Realistic MOEX Costs)	-12.59%	33.81%	-0.61	-0.31	-68.70%	-0.18
7. 1-Bar Momentum (0-Cost Academic)	-18.37%	33.79%	-0.80	-0.50	-74.47%	-0.25
8. 1-Bar Momentum (Realistic MOEX Costs)	-34.38%	33.82%	-1.44	-1.29	-91.53%	-0.38

2. €


MemLabs 'HFT Quant Trilogy' on MOEX (2021–2026)
1. FORTS Basis Arbitrage & Repo Carry
Institutional Adapted: Long-Only AR + Repo Cash (SBER)
MemLabs Adaptive Regime Switcher (SBER) (Vid 3)
2. Trend Following: SMA-200 + Repo Cash
MemLabs 1-Bar Momentum (SBER) (Academic 0-Cost)
Benchmark: Buy & Hold SBER
Benchmark: Buy & Hold IMOEX
MemLabs 1-Bar Mean Reversion (SBER) (Academic 0-Cost)
MemLabs 1-Bar Momentum (SBER) (Realistic MOEX Costs)
MemLabs 1-Bar Mean Reversion (SBER) (Realistic MOEX Costs)
Benchmark: Risk-Free (Key Rate)
Equity Multiplier (Base = 1.0)
2.0
1.5
1.0
0.5
0.0
Underwater Drawdown Profiles (%)
Drawdown (%)
0
-20
-40
-60
-80
2021-01
2022-01
2023-01
2024-01
2025-01
2026-01
2027-01
Date

3. • , , MemLabs f

3.1. „ 1-Bar Momentum (Trend Following) MOEX?

‘ 5tyLyYY1PXY … … … Š … 1: $s_t = \text{sign}(r_{t-1})$. ’
Œ …f Š Œ …† Š …ŠŒ Š (… …†
…). „ ‹ ‹ ‰
(-18.37% CAGR). ‘ …f 0.05% ‹ (25%)
… ‰ f : -34.38% CAGR -91.53%.

3.2. „ Mean Reversion: , … , , †

‘ _urG139AM48 …: $s_t = -\text{sign}(r_{t-1})$. ‘ … ‰
…† +9.09% ( ).
‹ † … ‰ “ Š, …‰ … † Œ 25%
. „… ‹ ‹ “ , “ Š Œ …† …f
-12.59% -68.70%.

3.3. Regime Changes (X5QcNyYRMqQ) f f

‘ † † Œ ‰ :
” ‘ † (… …† † < 22%) …† ;
” ‘ (22% < Vol < 35%) ;
” ‘ (Vol > 35% … ‰ SMA-200) • 100% ‹ - Ž„ €•.
…† : Š …† … CAGR +10.27% -20.80%,
Œ …† Long-Only AR (|y| > 0.15%) … CAGR +13.42%
-22.02%!

4. ‡ f , , ,

1. • …† - …† Š (75% …): FORTS Basis Carry & Ž„ €,
” – / ˆ ‰ , + 1.0%
(14.4%) .
” “ – ‰ (LQDT / RUSFAR) ‰
… Œ €• .

2. •…† - ..... (25% …): • …f …† ‰ (Regime Switcher) + Long-Only AR
” ‘ … … (, ˜ …) ‹ SMA-200 ‰
… . „ … …† (>35%) • … ‰ .
” • † ..... : 13.4% …† .

3. – : 1- , … -‹
Œ Š .